COMPUTER ENGINEERING

QUEZON CITY

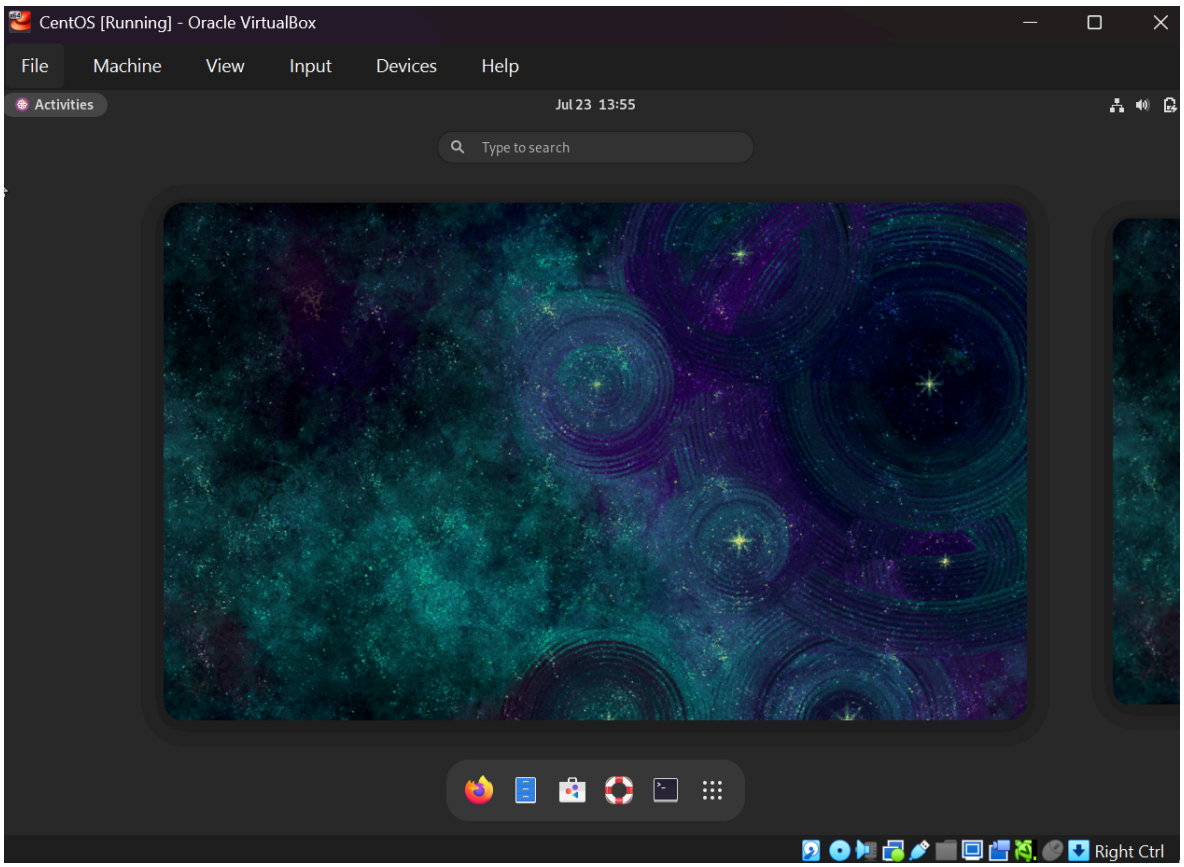
COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	7/23/2026	**DATE SUBMITTED:**	7/23/2026
NAME:	Alvarez, Erick Jefferson A.	**INSTRUCTOR:**	Engr. Quejado

ACTIVITY NO. 3.1 Install SSH server on CentOS or RHEL 8

1. PROCEDURE OUTPUT

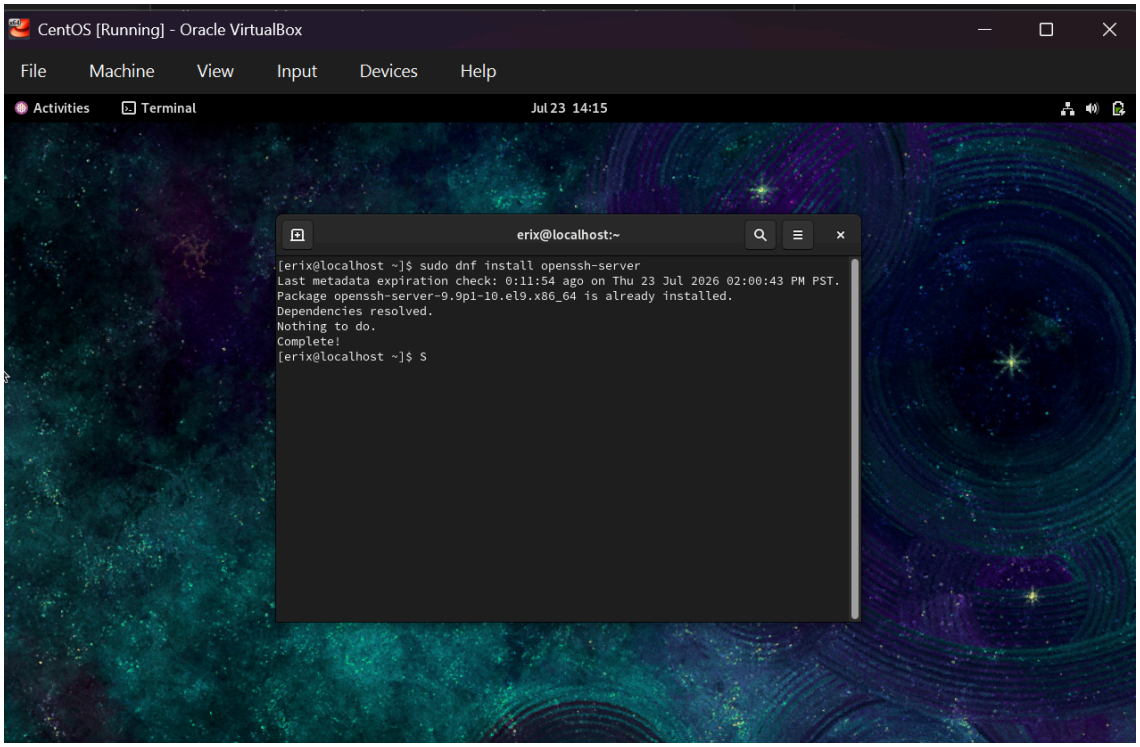
Task 1: Download the CentOS or RHEL-8 image (Create screenshots of the following)

1. Download the image of the CentOS here:
http://mirror.rise.ph/centos/7.9.2009/isos/x86_64/
2. Create a VM machine with 2 Gb RAM and 20 Gb HD.
3. Install the downloaded image.
4. Show evidence that the OS was installed already.

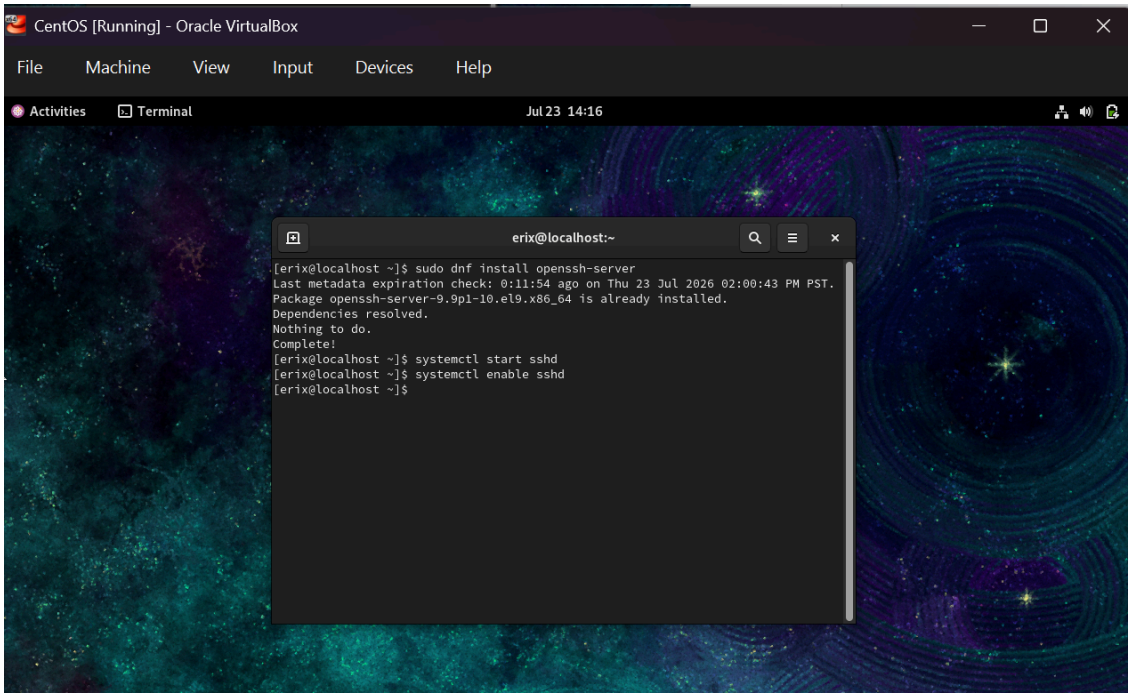


Task 2: Install the SSH server package openssh

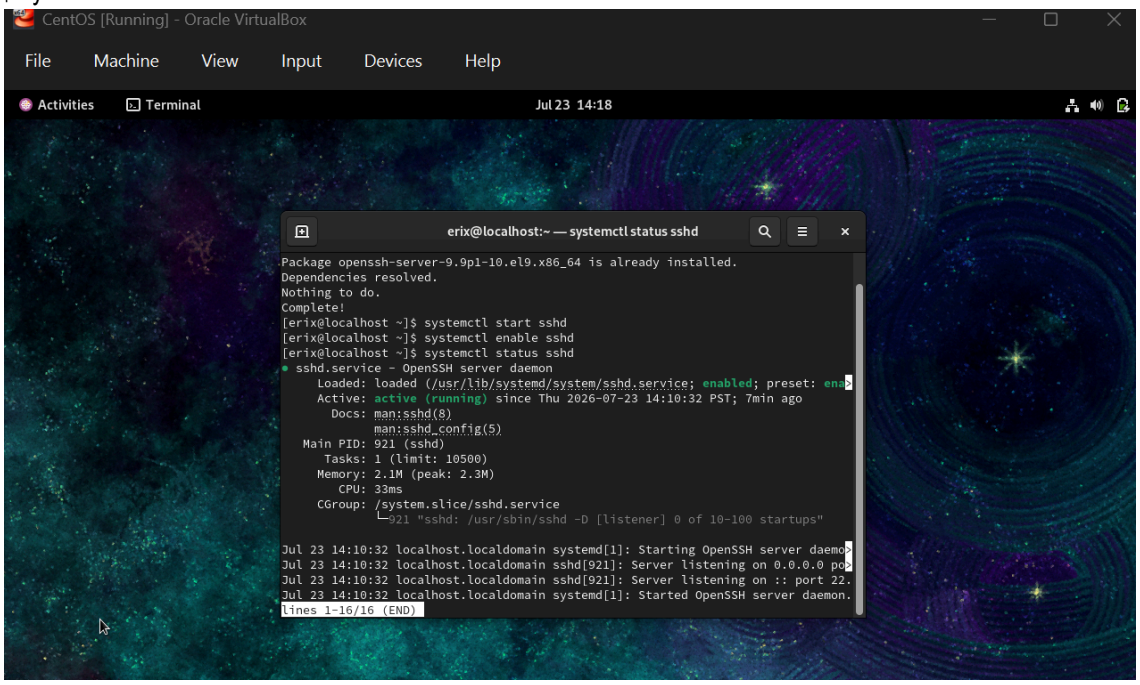
1. Install the ssh server package openssh by using the dnf command:
$ dnf install openssh-server



2. Start the sshd daemon and set to start after reboot:
$ systemctl start sshd
$ systemctl enable sshd



3. Confirm that the sshd daemon is up and running:
$ systemctl status sshd



4. Open the SSH port 22 to allow incoming traffic:
$ firewall-cmd --zone=public --permanent --add-service=ssh
$ firewall-cmd --reload

5. Locate the ssh server man config file /etc/ssh/sshd_config and perform custom configuration. Every time you make any change to the /etc/ssh/sshd-config configuration file reload the sshd service to apply changes:

```
#PermitRootLogin prohibit-pas
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication no
```

- I changed the pubkey authentication to no

$ systemctl reload sshd

```
[erix@localhost ~]$ ^C
[erix@localhost ~]$ ^C
[erix@localhost ~]$ systemctl reload sshd
[erix@localhost ~]$ S
```

Task 3: Copy the Public Key to CentOS

1. Make sure that ssh is installed on the local machine.
2. Using the command ssh-copy-id, connect your local machine to CentOS.

```
erix@Ubuntu:~$ ssh-copy-id erix@192.168.56.101
The authenticity of host '192.168.56.101 (192.168.56.101)' can't be established
ED25519 key fingerprint is SHA256:4Cvm1ZP29r7fFcezHYriiaslgFx0IGr3J5SL1pNHK+0.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
```

3. On CentOS, verify that you have the authorized_keys.

```
[erix@localhost ~]$ cat ~/.ssh/authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIOnfvbhPdwqWQ4oFMzCydU8V9JcbjWyTYBTUscvp3a+G erix@Ubuntu
[erix@localhost ~]$ S
```

Task 4: Verify ssh remote connection

1. Using your local machine, connect to CentOS using ssh.
2. Show evidence that you are connected.

```
erix@Ubuntu:~$ ssh erix@192.168.56.101
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Jul 23 17:37:27 2026
[erix@localhost ~]$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?

- The choice of the perfect operating system will depend on your usage. Different distributions have their different applications. While some distributions are simple to install and easy to use, others tend to be more difficult to configure. It would be necessary to establish whether a distribution is user friendly. Additionally, some distributions come free, such as the Debian distribution, while others need you to pay a subscription fee, for example Red Hat. Therefore, we have to consider all these aspects when selecting the appropriate one.

2. What are the main differences between Debian and Red Hat Linux distributions?

- Both Debian and Red Hat are examples of Linux distributions, which aim at satisfying different needs of users. Debian is an open-source OS aimed at providing stability to its users. This distribution is often installed on personal computers and servers. Red Hat focuses more on the needs of businesses, providing them with professional support, high levels of security, and commercial services. The difference is also in the management of software packages: Debian uses the APT tool, and Red Hat works with RPM/DNF.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **chatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Troubleshoot and fix errors**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **ChatGPT was used for troubleshooting. All work was completed and verified by me**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.



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